

Alex S. Huang

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EDUCATION

♦ **University of Texas at Dallas, Erik Jonsson School of Engineering** **Expected May 2028**

Bachelor of Science in Computer Engineering,

- Cumulative GPA: 4.0/4.0 | Attending on full-ride merit scholarship | Hobson Wildenthal Honors College

♦ **Hamilton High School | Chandler, AZ** **May 2025**

- Weighted GPA: 4.8/4.0 | Summa Cum Laude | National Merit Scholar | ISEF International Finalist

EXPERIENCE

♦ **University of Texas at Dallas | Richardson, TX** **Aug. 2025 – Present**

◇ **Intelligent Robotics and Vision Lab, *Robotics Research Assistant***

- Constructing and fine-tuning **Koch v1.1 robotic grippers** for **smart policy-learning** and **task completion**.
- Assist in original research in robot perception, planning, control and learning under Prof. Xiang daily in the lab.

♦ **Arizona State University | Tempe, AZ** **Jun. 2024 – Jul. 2025**

◇ **Sun Robotics Lab, *Intelligent Robot Dynamics & Actuators Intern*** *Sep. 2024 – Aug. 2025*

- Engineered **AI segmentation pipeline** of real-time finger-angle classification for **robotic hand teleoperation** leveraging **DeepLabV3, MobileNetV2, and openCV**, **reducing costs by 90%** compared to existing solutions.
- Refined robotic hand (DexHand) CAD design with **custom 3D-parts, soldering, and actuator** modifications
- Selected as **ISEF 2025 National Finalist** to represent Arizona, accepted into **2025 IEEE ICSECS conference**

◇ **Security Engineering for Computing Lab, *Cybersecurity/SWE Intern*** *Jun. 2024 – Aug. 2024*

- Built **SmishSmashing**, a GPT4-powered extension designed to counteract smishing attacks; integrates **Twilio** for SMS parsing, **MongoDB** for storage, and **AWS EC2** for deployment. Improved **smishing detection** by 80%.

♦ **Stanford University | Palo Alto, CA** **Jun. 2024 – Aug. 2024**

◇ **Stanford Compression Forum (SCF), *STEM2SHTEM Summer Research Intern***

- Selected as **1/30 high school students worldwide** to attend **STEM/AI research** led by Stanford professors.
- Co-authored paper analyzing **prompt engineering** tasks for better **LLM decision-making** w/ **Prof. Ben Roy**.

♦ **Taiwan Overseas Community Affairs Council | Nantou, Taiwan** **Jul. 2023 – Aug. 2023**

◇ **English Teaching Service Program for Overseas Youth, *English Teacher/Volunteer***

- **Drafted & taught 4 weeks** of bilingual, hands-on English, STEM & arts curricula to rural grade school.

ACTIVITIES

♦ **Accessible Prosthetics Initiative - UTD Chapter** **Sep. 2025 – Present**

Electronic Arm Team Member

- Prototyping robotic arm, PCB boards, and **custom EMG w/ 50%+ performance** for the 2025 Dallas Abilities Expo.

♦ **Robotics - Hamilton High Microbots Team 698** **Aug. 2021 – May 2025**

Head of Marketing Operations, Secretary, Mechanic

- **Fundraised \$15,000+** w/ marketing team annually. Collaboratively **design & assemble** cost-efficient competition robots

SKILLS & INTERESTS

Programming: Python, C++, JavaScript, HTML, Bash

Software Tools: GitHub, API integration, MongoDB, AWS, machine learning, Canva, MS Office, Google Suite

Hardware: Shop machinery and power tools, soldering, Arduino, 3D printing/assembly, CAD (Fusion/OnShape)

Languages: English (native), Mandarin (native), Spanish

Interests: Anything robotics, PC building, chamber music, National Parks, cooking, fantasy football, pickleball

AWARDS & HONORS

♦ **ISEF National Finalist & AzSEF Best of Fair**

Mar. 2025

♦ **UT Dallas National Merit Scholarship Recipient**

Apr. 2025

♦ **National Merit Finalist (top ~0.6% PSAT scorer)**

Feb. 2025

♦ **InnerView Service Ambassador Award (300+hrs)**

Jan. 2025